MIL-D-46205C (MU)
10 August 1966
SUPERSEDING
MIL-D-46205B (MU)
27 March 1964

MILITARY SPECIFICATION

DELAY OR NON-DELAY ELEMENT, M9
PARTS FOR

1. SCOPE

1.1 This specification covers quality assurance requirements for the parts for one type of delay element designated as Delay or Non-Delay Element, M9 (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-D-45608 - Designs and Drawings, Ordnance Inspection Equipment
MIL-C-45662 - Calibration of Standards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8838189 - Baffle
8838190 - Cup, delay
8838192 - Holder, primer
8838196 - Body
8863340 - Washer
8885300 - Holder, primer, non-delay

PUBLICATIONS

ORDNANCE CORPS

IEL-8816502 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material shall be in accordance with applicable drawings and specifications

3.2 Parts.-Parts shall comply with all requirements specified on applicable drawings and in applicable specifications.

3.3 Workmanship.-All manufacturing and processing shall be in a thorough workmanlike manner. Parts shall be free from cracks, sharp corners where not required, burrs, chips, dirt, grease, rust, or other foreign material. The cleaning method shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection.

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit an first article sample to a Government approved facility as designated by the Contracting Officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 20 each of every component which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

4.3.2.1 Body (see dwg. 8838196).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Outside diameter	Gage	01001
102.	Location of delay or non-delay assembly hole	Gage	01002
103.	Diameter of delay cup cavity	Gage	01003
104.	Diameter of primer holder cavity	Gage	01004
105.	Concentricity of primer holder cavity with delay cup cavity	Gage	01005
106.	Location of guide pin slot	Gage	01006
107.	Width of guide pin slot	Gage	01007
108.	Location of lock pin slot	Gage	01008
109.	Distance to shoulder of lock pin slot	Gage	01009
110.	Crack or split	Visual	01010
Minor:	AQL 2.5 percent		
201.	Total length	Gage	01011
202.	Distance from body groove at lock pin slot to guide pin slot	Gage	01012
203.	Protective coating damaged exposing base metal	Visual	01013
204.	Chamfer missing	Visual	01014
205.	Evidence of poor workmanship (see 3.3)	Visual	01015

4.3.2.2. Holder, primer (see dwg. 8838192).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

Major:	AQL 1.0 percent		
101.	Pitch diameter of internal thread	Gage	02001
102.	Minor diameter of internal thread	Gage	02002
103.	Length of full internal thread, min	Gage	02003
104.	Diameter of primer cavity	Gage	02004
105.	Depth of primer cavity	Gage	02005
106.	Profile and height of crimping lip	Gage	02006
107.	Diameter of counterbore at bottom of delay cup cavity	Gage	02007
108.	Depth of delay cup cavity	Gage	02008
109.	Concentricity of pitch diameter of thread with outside diameter	Gage	02009
110.	Diameter of flash hole	Gage	02010
111.	Crack or split	Visual	02011
Minor:	AQL 2.5 percent		
201.	Outside diameter, maximum (max)	Gage	02012
202.	Total length	Gage	02013
203.	Outside diameter of crimping lip	Gage	02014
204.	Depth to bottom surface of counterbore below delay cup assembly cavity	Gage	02015
205.	Chamfer missing	Visual	02016
206.	Countersink missing	Visual	02017
207.	Protective coating damaged exposing base metal	Visual	02018
208.	Evidence of poor workmanship (see 3.3)	Visual	02019

4.3.2.3 Holder, primer, non-delay (see dwg. 8885300).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Diameter of relay cavity	Gage	03001
102.	Depth of relay cavity	Gage	03002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

103.	Length of small outside diameter	Gage	03003
104.	Diameter of primer cavity	Gage	03004
105.	Depth of primer cavity	Gage	03005
106.	Profile and height of primer crimping lip	Gage	03006
Minor:	AQL 2.50 percent		
201.	Diameter of flash hole, max	Gage	03007
202.	Small outside diameter at relay end	Gage	03008
203.	Large outside diameter at relay end	Gage	03009
204.	Length from shoulder at rear of larger outside diameter to forward surface	Gage	03010
205.	Large outside diameter	Gage	03011
206.	Evidence of poor workmanship (see 3.3)	Visual	03012

4.3.2.4 Cup, delay (see dwg. 8838190).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split	Visual	04001
Major:	AQL 1.0 percent		
101.	Diameter of charge cavity, min.	Gage	04002
102.	Diameter of flash hole, min.	Gage	04003
103.	Outside diameter forward of thread, max	Gage	04004
104.	Pitch diameter of thread	Gage	04005
105.	Major diameter of thread	Gage	04006
106.	Length of full thread, min.	Gage	04007
107.	Concentricity of flash hole with pitch diameter of thread	Gage	04008
108.	Finish improper on cavity diameter	Visual	04009
Minor:	AQL 2.5 percent		
201.	Diameter of counterbore, min.	Gage	04010
202.	Depth to bottom of cavity	Gage	04011
203.	Depth of counterbore	Gage	04012

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

204.	Protective coating damaged exposing base metal	Visual	04013
205.	Chamfer missing	Visual	04014
206.	Evidence of poor workmanship (see 3.3)	Visual	04015

4.3.2.5 Baffle (see dwg. 8838189).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter, min	Gage	05001
102.	Diameter of flash hole	Gage	05002
103.	Location of flash holes with outside diameter	Gage	05003
Minor:	AQL 2.5 percent		
201.	Outside diameter, max	Gage	05004
202.	Thickness	Gage	05005
203.	Protective coating damaged exposing base metal	Visual	05006
204.	Evidence of poor workmanship (see 3.3)	Visual	05007
205.	Radius excessive	Visual	05008

4.3.2.6 Washer (see dwg. 8863340).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Diameter of hole, min	Gage	06001
102.	Outside diameter, min	Gage	06002
103.	Thickness at bottom	Gage	06003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

Minor:		AQL 2.5 percent	
201.	Thickness at flange	Gage	06004
202.	Total height	Gage	06005
203.	Outside diameter, max	Gage	06006
204.	Profile and angle of top	Gage	06007
205.	Evidence of poor workmanship (see 3.3)	Visual	06008

4.3.3 Testing

4.3.3.1 Salt spray test - Minor defect, for Body dwg. 8838196, Code No. 01016, Holder, primer dwg. 8838192, Code No. 02019, Holder, primer, non-delay dwg. 8885300, Code No. 03012, Cup, delay dwg. 8838190, Code No. 04016 and Baffle dwg. 8838189, Code No. 05010. Samples shall be selected or panels prepared for each of the above parts (as applicable) and tested in accordance with the specification as required by the applicable drawing. Any sample part or panel, failing to comply with the applicable requirements shall be cause for rejection of the lot. Action shall be taken as specified in the applicable specification.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-8816502 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

formance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3.4.4 Calibration.-The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with Specification MIL-C-45622.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.

5.1.1 Level C.-The metal parts shall be packaged, packed and marked in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Distribution of data cards shall be as specified by the procuring activity.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflcted on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment tht has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contrator. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.

or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Intended use.-It is intended that the items of this specification, when loaded, will be used with Bomb Fuzes M904E2 and M905.

6.5 Submission of first article sample.-Instructions, as to the location for the evaluation of the first article sample, shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Cudtodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A181

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:12Z
Check the source to verify that this is the current version before use.